

Application of Stochastic Optimal Control in Finance.

Objective of project:

To study the general theory of stochastic control and optimization. This is a sub-field of classical control theory that deals with subjection to random perturbations that drive a dynamical system. The system can be controlled in order to find a optimum for some performance criterion. The aim is then to design a path through time for the controlled variables that performs the desired control task which is found in a performance criterion.

Basically treat a stochastic control model where the state of the system is governed by the following controlled SDE valued in $\mathbb{R}^n$,

$$dX_s = \mu(s, X_s, \alpha_s)ds + \sigma(s, X_s, \alpha_s)dW,$$

The control process $\alpha = (\alpha_s)$, $s \geq 0$ is a progressively measurable process, in a subset of $\mathbb{R}^m$.

Let the gain functional under suitable assumptions be defined as

$$J(t, x, \alpha) = E \left[\int_t^T f(s, X_s^{t,x}, \alpha_s) ds + g(X_T^{t,x}) \right]$$

Having stated the basis of the control problem, we are going to look at two different but intertwined approaches to find the optimal control and corresponding value function.

Methodology:

Approach one: Hamilton Jacobi Bellman approach: A result of the dynamic programming principle is the Hamilton-Jacobi-Bellman equation (HJB). The HJB describes the local behavior of the value function when the stopping time θ goes to t . In optimal control theory, it will give us necessary conditions for optimality of a control with respect to the driver of the functional. Given a smooth solution to the HJB equation, the candidate coincides with the value function. Many a times the HJB is nonlinear degenerate PDE so we might have to look for viscosity solution.

Furthermore, we are interested in the link between the HJB solution and BSDEs. As we have noticed, the HJB equation is a form of non-linear PDE. We look at the extension of the Feynman-Kac theorem for semilinear PDE. We shall represent the solution to the PDE by means of the BSDE. There's a theorem in [1] that will show that a classical solution to the semilinear PDE provides a solution to the BSDE.

We will exploit this relationship when utilizing a numerical approach to approximate the solution of BSDEs and determine the optimal control for the corresponding stochastic control problem

Approach Two: The Maximum Principle Approach: It states that an optimal state trajectory must solve a Hamilton system together with a maximum condition of the generalized Hamiltonian. The maximum principle establishes a crucial requirement that must be satisfied along an optimal trajectory. In contrast, the HJB approach computes a general value function that corresponds to the application of the optimal action, denoted as $\hat{\alpha}$. This computation eliminates the optimization in the Hamiltonian, but it also forfeits the global properties associated with the HJB equation. SMP optimizes along a trajectory, as opposed to the value function $v(t, x)$ over the whole state space. Therefore, it can only be at most local optimal in the space of possible trajectories.

The end result is a FBSDE system with a coupling determined by the feedback mapping. In most cases, especially in situations which are not designed as toy models, we get a fully coupled FBSDE. In

general there is no available explicit solution to this system. However in [2] they propose an extension of the BCOS method for coupled forward-backwards stochastic differential equations. This we can use for our gain to solve complex stochastic control problems.

Potential Application to Finance:

Portfolio allocation, Production-consumption model, Irreversible investment model, Quadratic hedging of options, Optimal selling of an asset, Valuation of natural resources etc.

Bibliography:

[1] Huy^{en} Pham. Continuous-time stochastic control and optimization with financial applications, volume 61. Springer Science & Business Media, 2009.

[2] Thomas P Huijskens, Maria J Ruijter, and Cornelis W Oosterlee. Efficient numerical fourier methods for coupled forward–backward sdes. *Journal of Computational and Applied Mathematics*, 296:593–612, 2016.